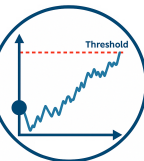


Intro to EAM

@47th ECVP 2025 Mainz

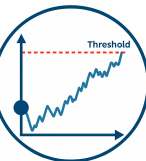
Margherita Calderan

August 24, 2025



Perception into action

Evidence Accumulation Models assume that, upon stimulus presentation, the decision maker:

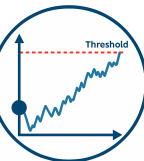


What is “evidence”?

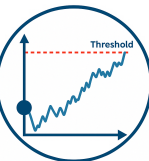
Neural signals reflecting sensory or internal information relevant to a choice.

“Are dots moving to the right or to the left?”

- Neurons fire in proportion to motion direction & strength.
- These firing rates = **momentary evidence**.

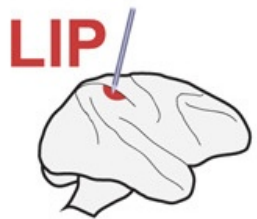


How does the brain accumulate evidence?

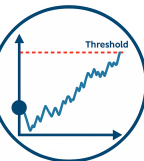
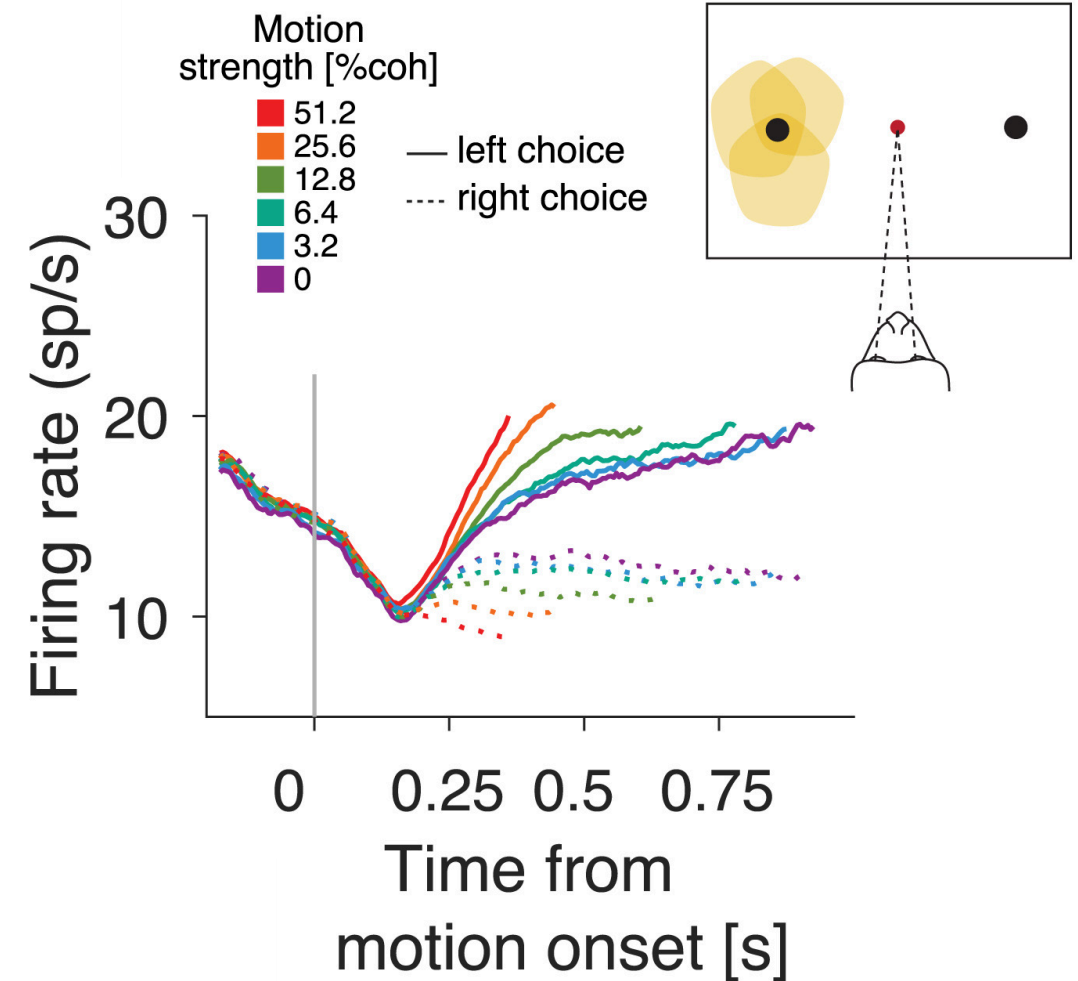


Integrator neurons receive this input

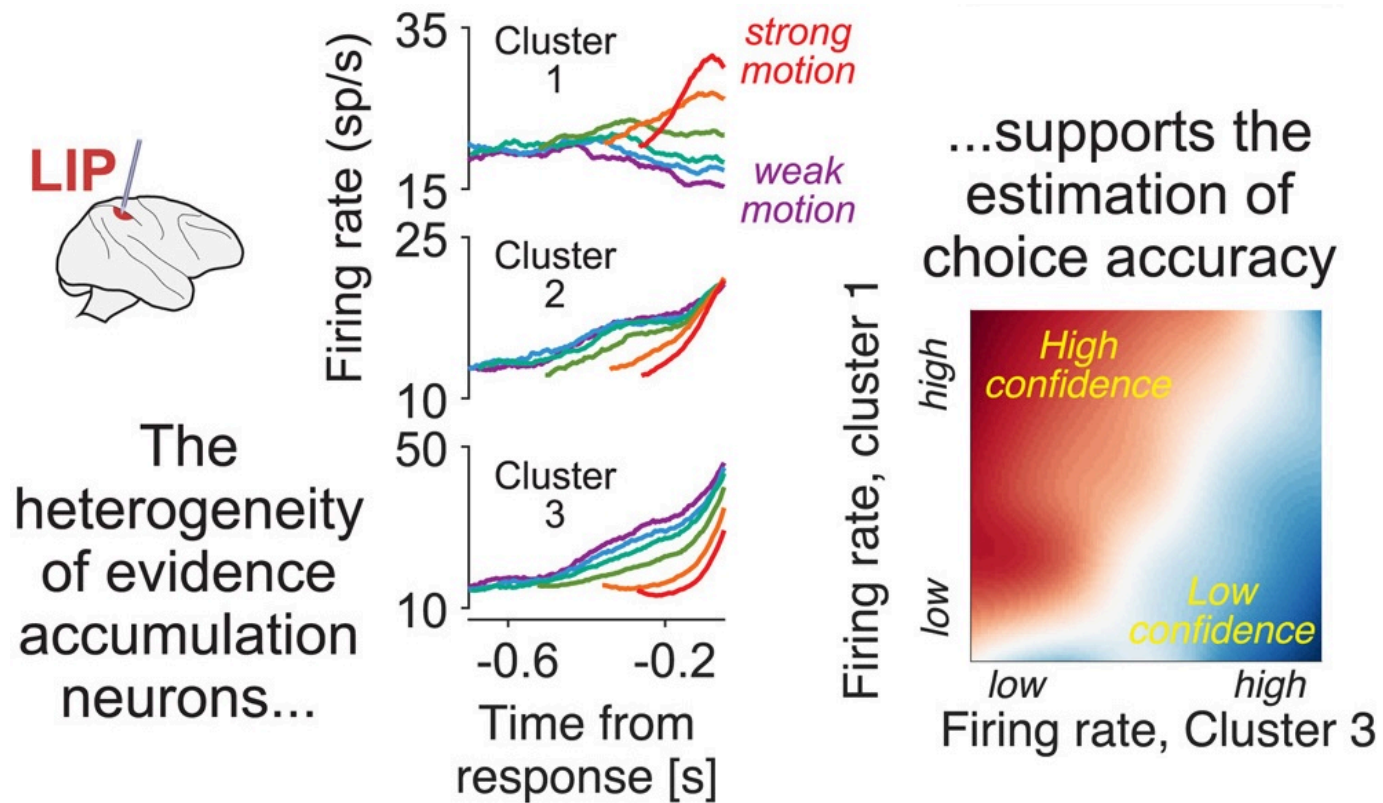
- Neurons in **LIP**, **dIPFC**, or **striatum** ramp up/down over time.
- Reflects **accumulated evidence**



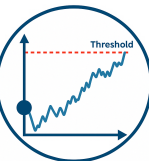
Zylberberg & Shadlen, 2025



Threshold crossing triggers a decision

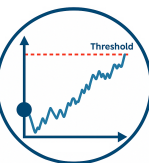


Zylberberg & Shadlen, 2025

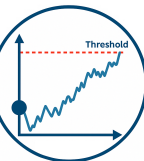


Brain computation of decisions

Cognitive Term	Neural Interpretation
<i>Evidence</i>	Sensory neuron firing rates
<i>Accumulation</i>	Integration in parietal/frontal areas
<i>Bound</i>	Decision threshold in firing/activity



Latent Cognitive Parameters

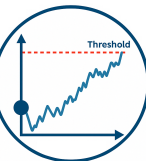


Sequential processing assumption

Total Response Time (RT) is modeled as the sum of **three sequential stages**:

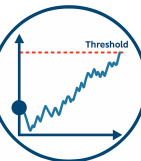
1. Stimulus encoding
2. Evidence accumulation (decision-making)
3. Motor response execution

Stages (1) and (3) are captured in the **nondecision time (T_{er})** parameter.



EAMs decompose decisions into:

- **Drift rate**
- **Threshold**

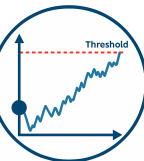


Drift Rate

Reflects **evidence strength**

- **↑** Drift: fast & accurate decisions
- **↓** Drift: slow, error-prone

Manipulated by stimulus discriminability/task difficulty



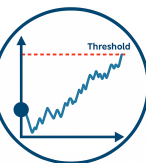
Thresholds

Set before stimulus onset

Reflect **response caution/cognitive control/bias/preference**

- **↑** Threshold: Slower but more accurate
- **↓** Threshold: Faster but error-prone

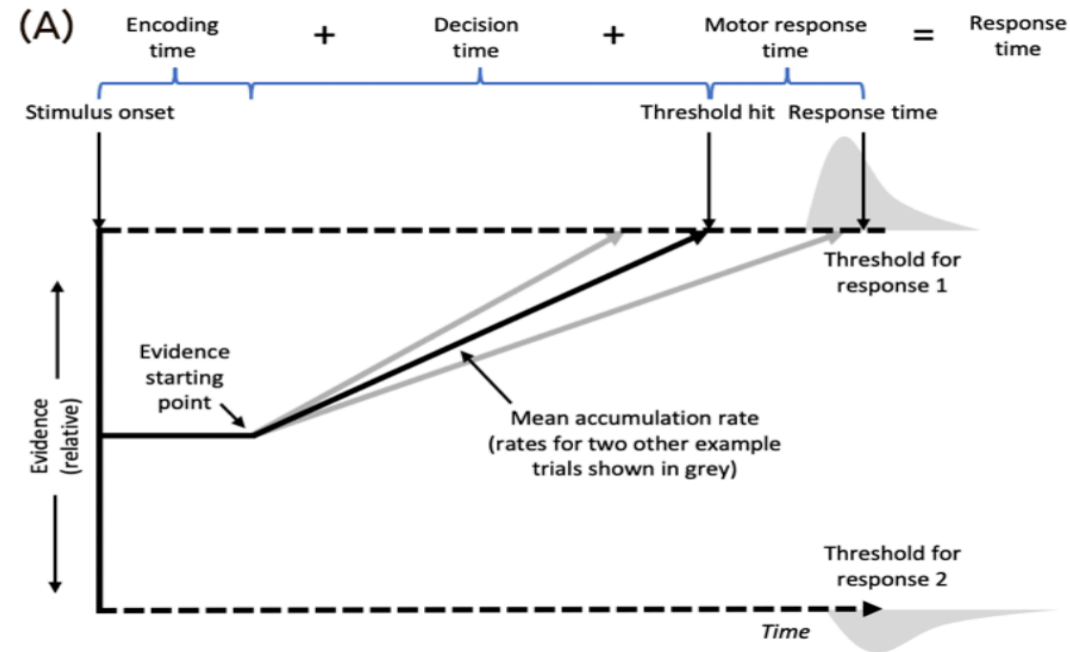
Manipulate via pre-trial cues or instructions.



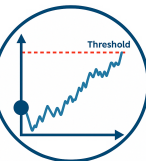
Relative Evidence Models

In relative evidence models (e.g., Wiener process, Diffusion Decision Model):

- Decision is based on the **difference** in accumulated evidence between two options.
- Suitable for **binary choices**.



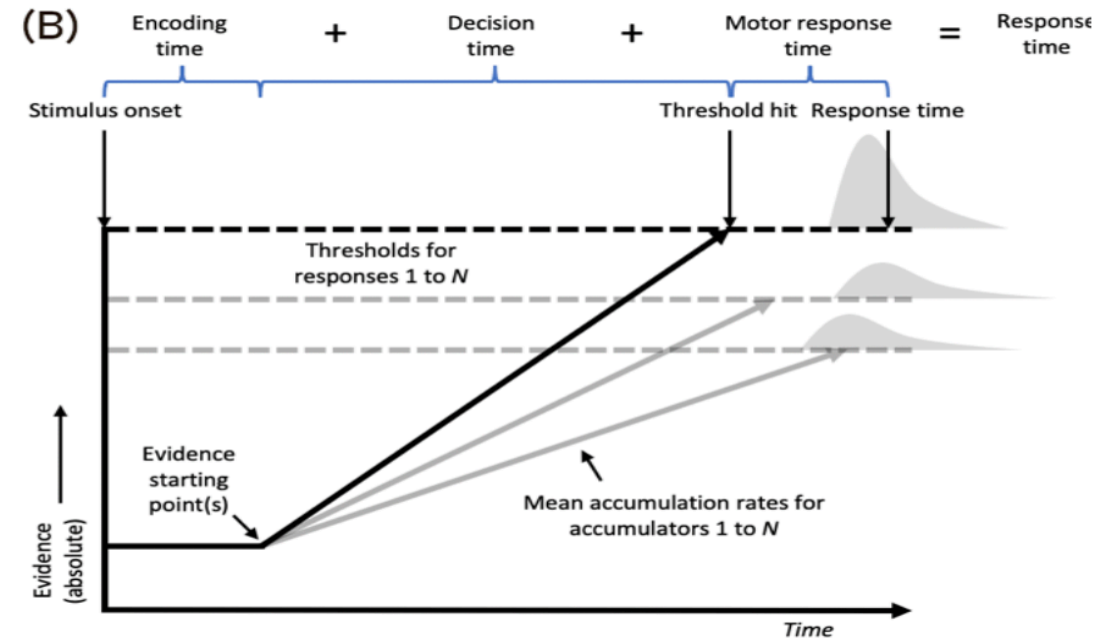
Boag, R. J., Innes, R. J., Stevenson, N., Bahg, G., Busemeyer, J. R., Cox, G. E., ... Forstmann, B. (2024, July 2). An expert guide to planning experimental tasks for evidence accumulation modelling.



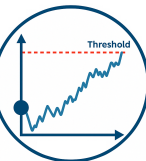
Absolute Evidence Models

In racing accumulator models (e.g., LBA, LNR, RDM):

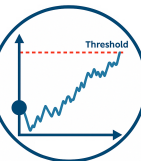
- Each option has its own **accumulator** tracking **absolute evidence**.
- Decision is made by the **first accumulator to reach threshold**.
- Can handle **multiple alternatives** (not just binary choices).



Boag, R. J., Innes, R. J., Stevenson, N., Bahg, G., Busemeyer, J. R., Cox, G. E., ...
 Forstmann, B. (2025) An expert guide to planning experimental tasks for evidence
 accumulation modelling.



Noisy process

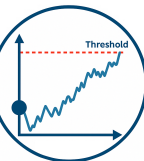
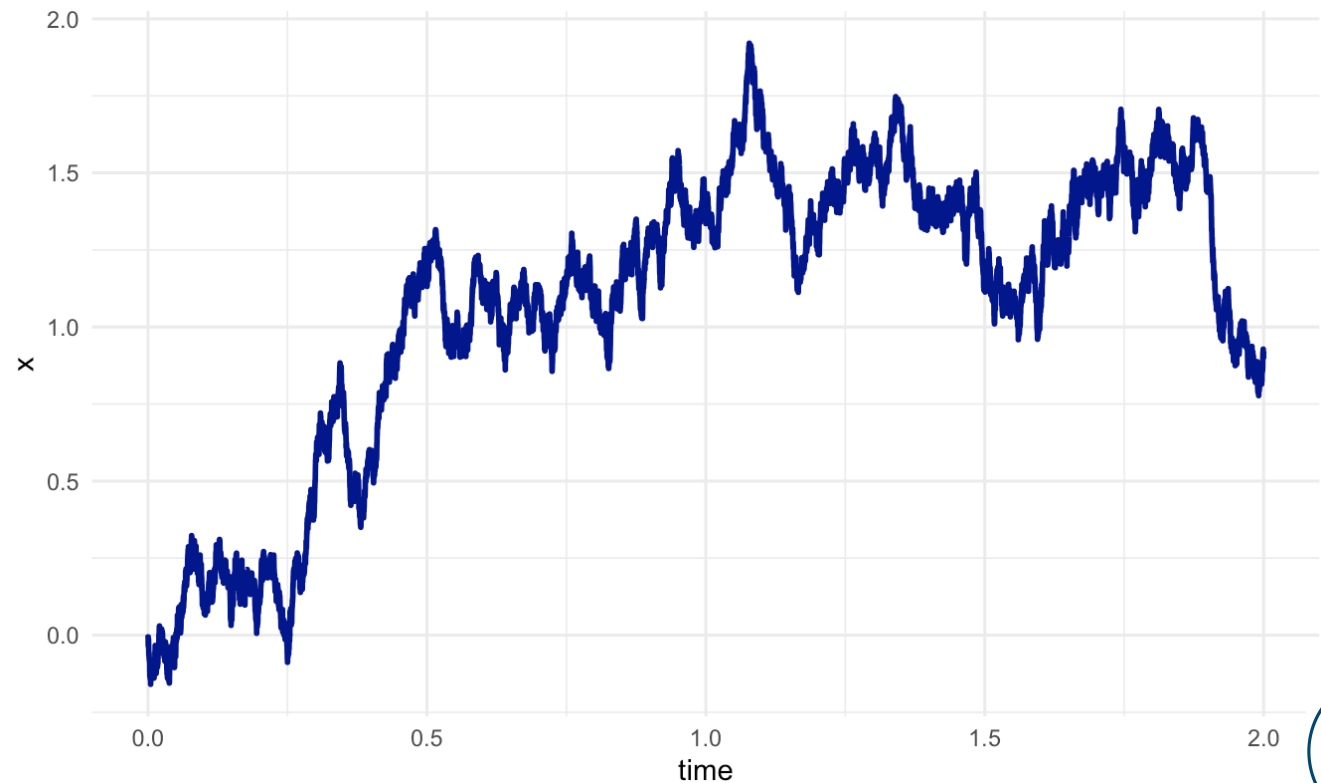


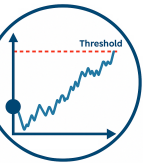
Wiener Process

The **Wiener process** (Brownian motion) is the foundation of many decision models ([Smith & Ratcliff, 2024](#)). It models the accumulation of evidence as a noisy process:

$$x(t) = z + vt + sW(t)$$

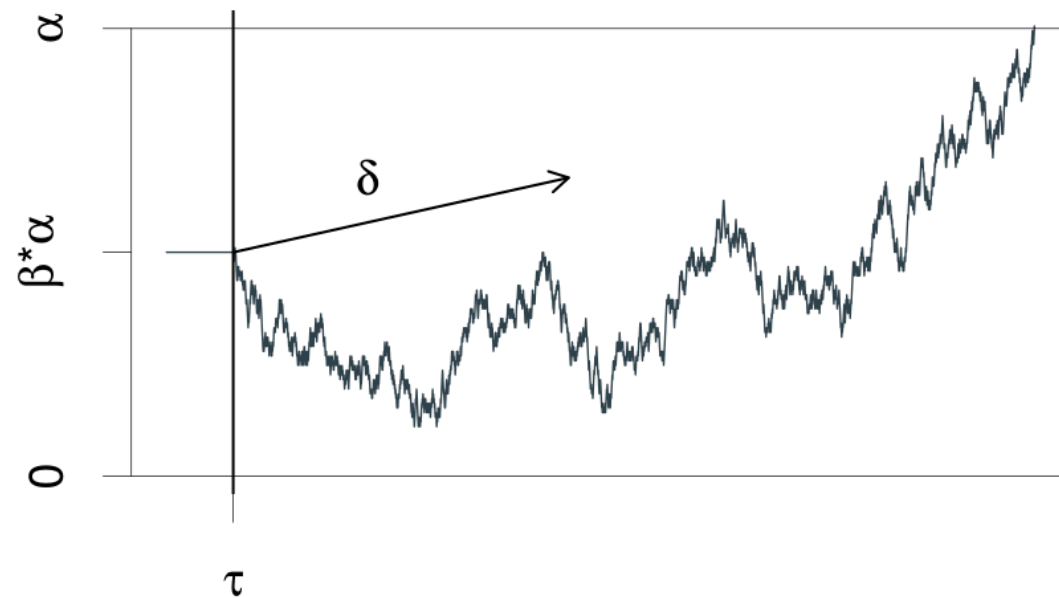
- $x(t)$: accumulated evidence at time t
- z : starting point
- v : drift rate (signal strength)
- s : diffusion coefficient
- $W(t)$: standard Wiener process (Gaussian increments)



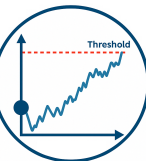


Wiener Diffusion Model

It is the expected distribution of the time until the process first hits or crosses one or the other boundary. This results in a bivariate distribution, over responses and hitting times.



- α : threshold
- β : initial bias (starting point)
- δ : quality of the stimulus (often v)
- τ : non-decision time



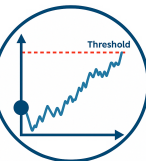
Full Diffusion Decision Model

The **Full DDM** accounts for more behavioral phenomena by allowing **trial-to-trial variability** in key parameters:

Drift rate

Starting point

Non-decision time



Full DDM Parameters

a : decision boundary

z: starting point

v : drift rate

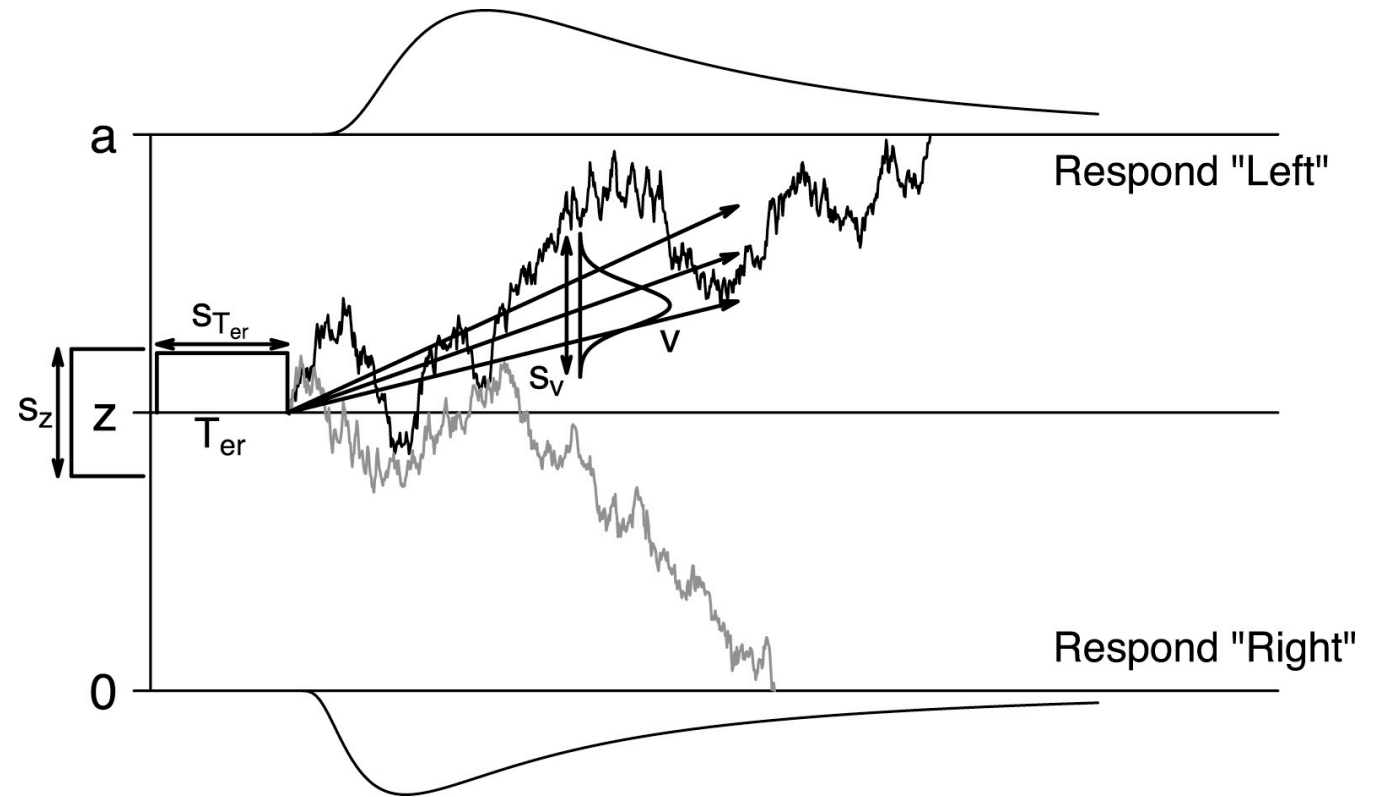
T_{er} : non-decision time

s : noise scale (usually fixed to 1)

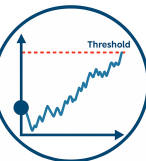
s_v : variability in drift rate

s_z : variability in start point

$s_{T_{er}}$: variability in non-decision time



Boehm, U., Annis, J., Frank, ... & Wagenmakers, E. J. (2018). Estimating across-trial variability parameters of the Diffusion Decision Model: Expert advice and recommendations. *Journal of Mathematical Psychology*, 87, 46-75. Ratcliff, R., & Rouder, J. N. (1998). Modeling Response Times for Two-Choice Decisions. *Psychological Science*, 9(5), 347-356.

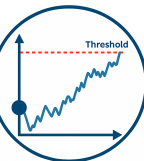


Purpose of variability

Adding variability improves the model's ability to:

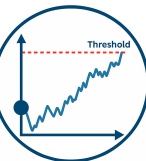
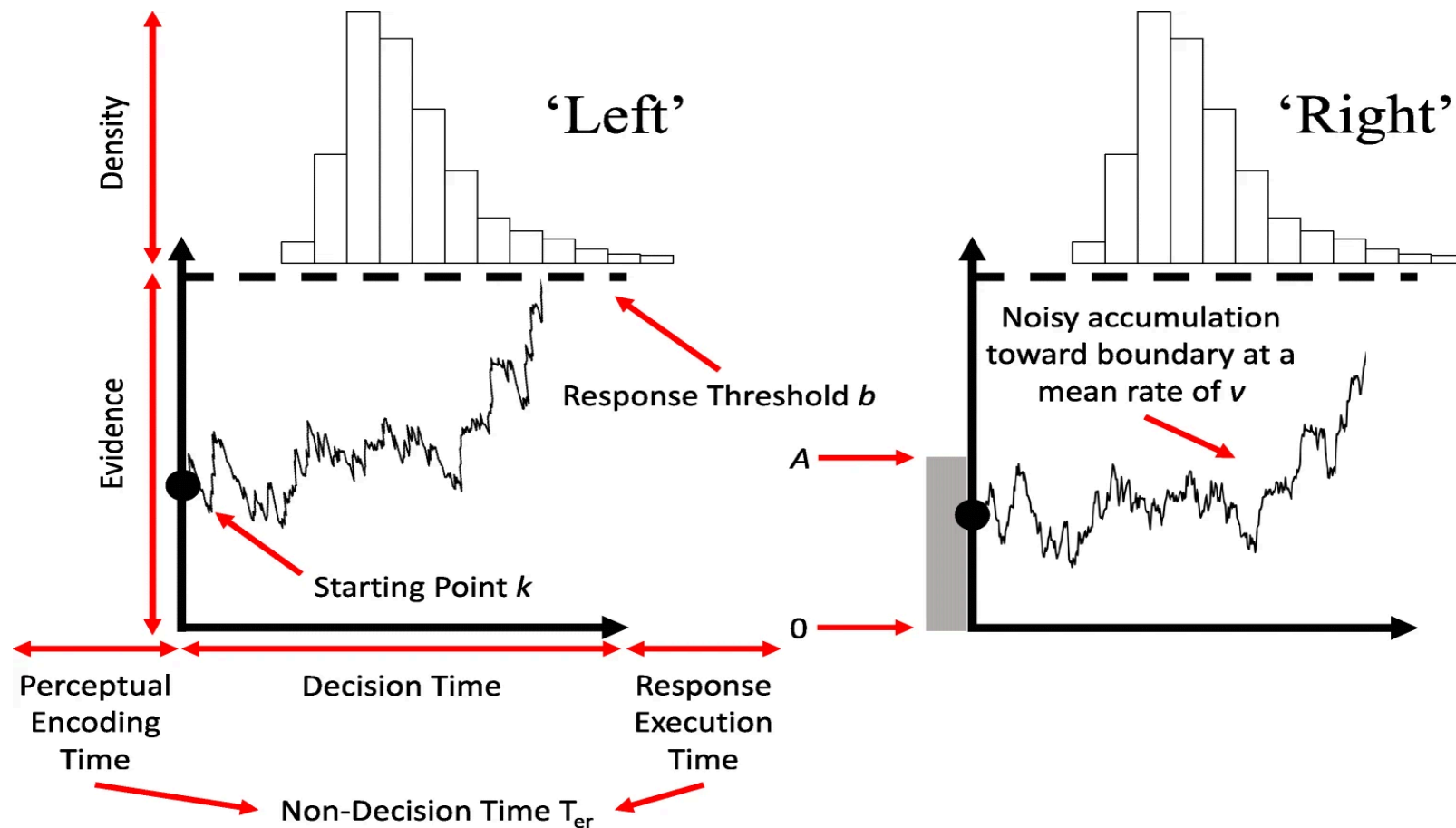
- Capture **error RT differences**
- Reflect **trial-to-trial attention or difficulty changes**

However, it increases computational demands.



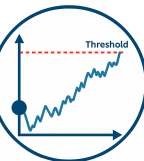
Racing Diffusion Model

Instead of a single process choosing between boundaries, the RDM uses **multiple independent diffusion processes**, one per option. Each accumulator races toward its threshold. The first to cross **wins**. [Tillman, Van Zandt, & Logan, 2020](#)



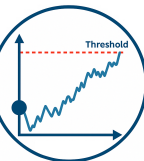
Summary

Model	Core Mechanism	Key Strengths
Wiener	Noisy accumulation	Simple FPT, binary outcomes
Full DDM	Accumulation + param variability	Realistic RTs, error patterns
Racing DM	Multiple accumulators	Handles multi-alternative decisions



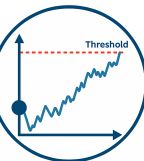
Core assumptions of the basic EAM

- Each decision = a **single, continuous accumulation** of evidence
- Culminates in a **discrete response**
- Evidence accumulates from **stimulus onset to response**



Guidelines

Boag et al. 2025



Within-Trial Stationarity

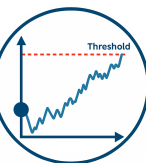
Model parameters are **fixed** within a trial

Evidence accumulation:

- Constant **mean rate**, though noisy
- No changes in stimulus evidence mid-trial

Decision thresholds:

- Set **before** stimulus onset
- **Do not change** during the trial

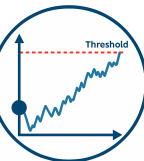


Within-Condition Stationarity

Parameters are constant **across same-type trials**

Assumes:

- Trials of same condition reflect **same cognitive settings**
- Participant behavior is stable



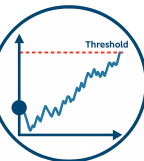
Free of contaminant processes

Data should reflect **evidence accumulation!**

Avoid:

- Random guessing
- Fast guesses
- Attention lapses or missing responses

Clean data = better model fit and interpretability.

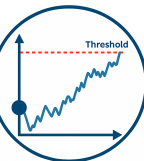


Response modality

Avoid imprecise, slow, or delayed responses.

Best:

- Manual keypresses
- Saccades



Response window

Starts with stimulus onset.

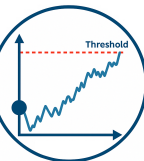
Ends with:

- A response
- Or a deadline

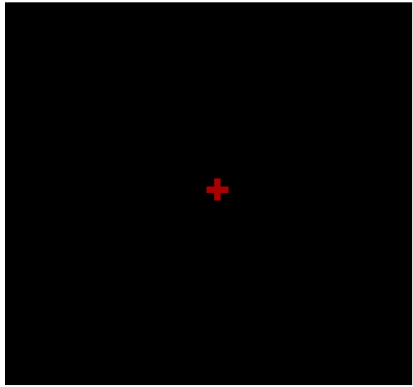
💡 Calibrate response window:

Long enough to allow natural responding

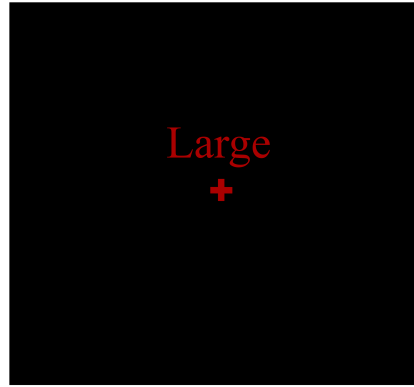
Short enough to avoid strategy shifts



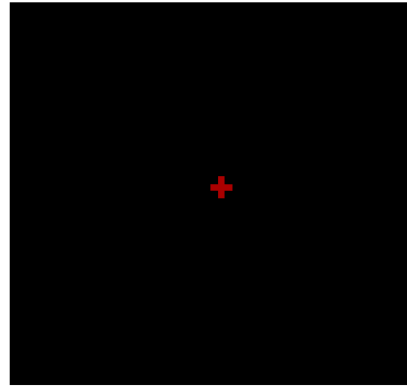
Practical Example (by Carolina Maria Oletto)



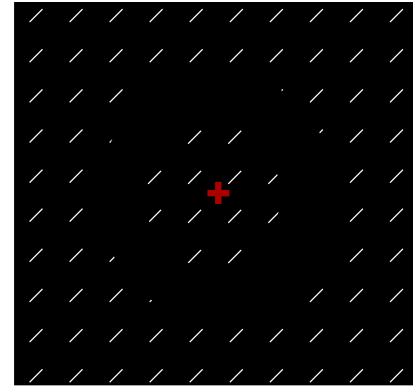
ISI (500-1000 ms)



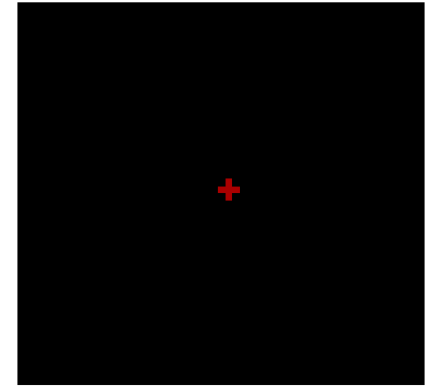
Cue (300 ms)



Fixation (300-500 ms)

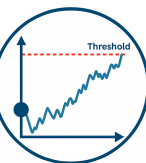


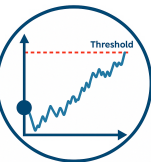
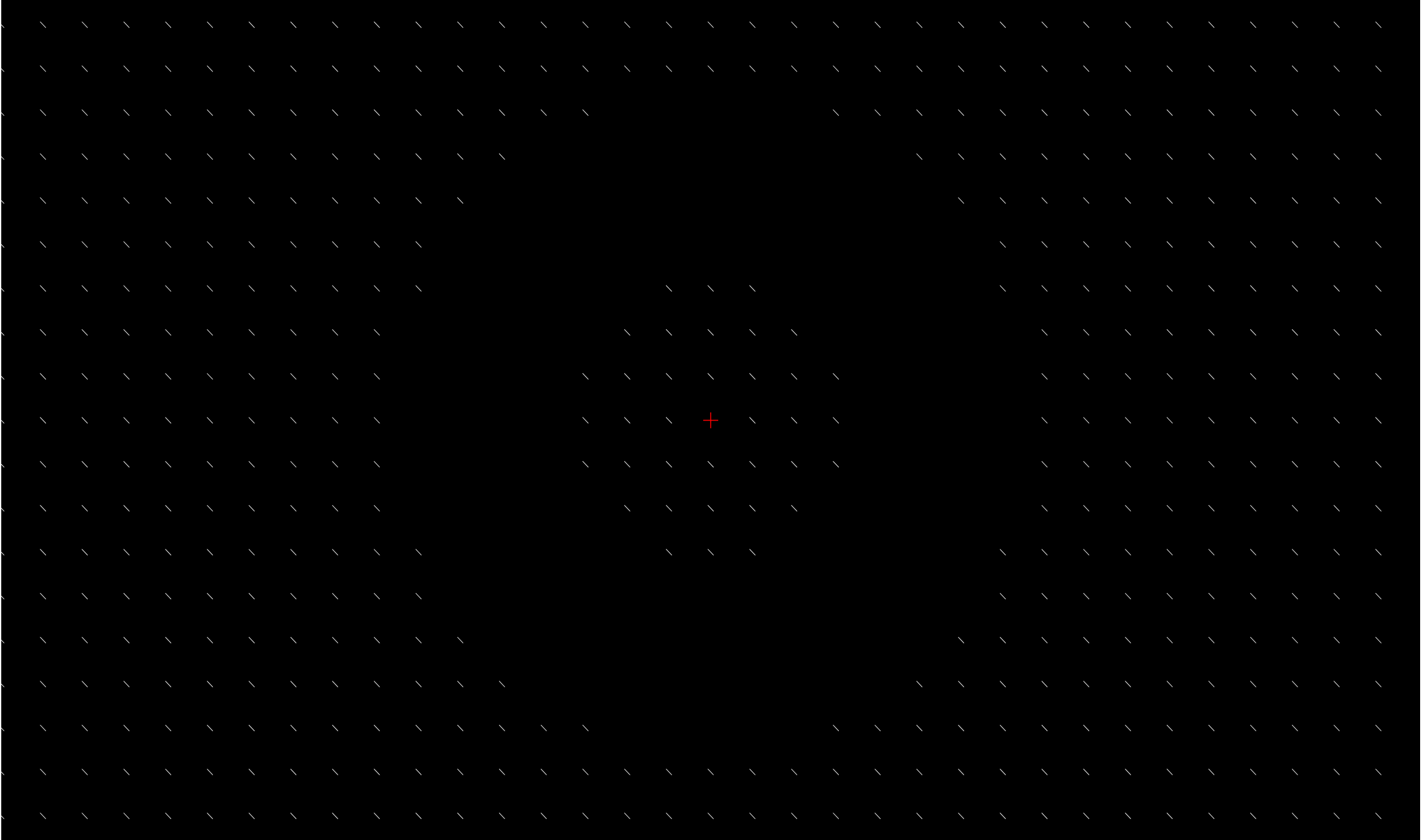
Stimulus (until response)

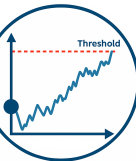
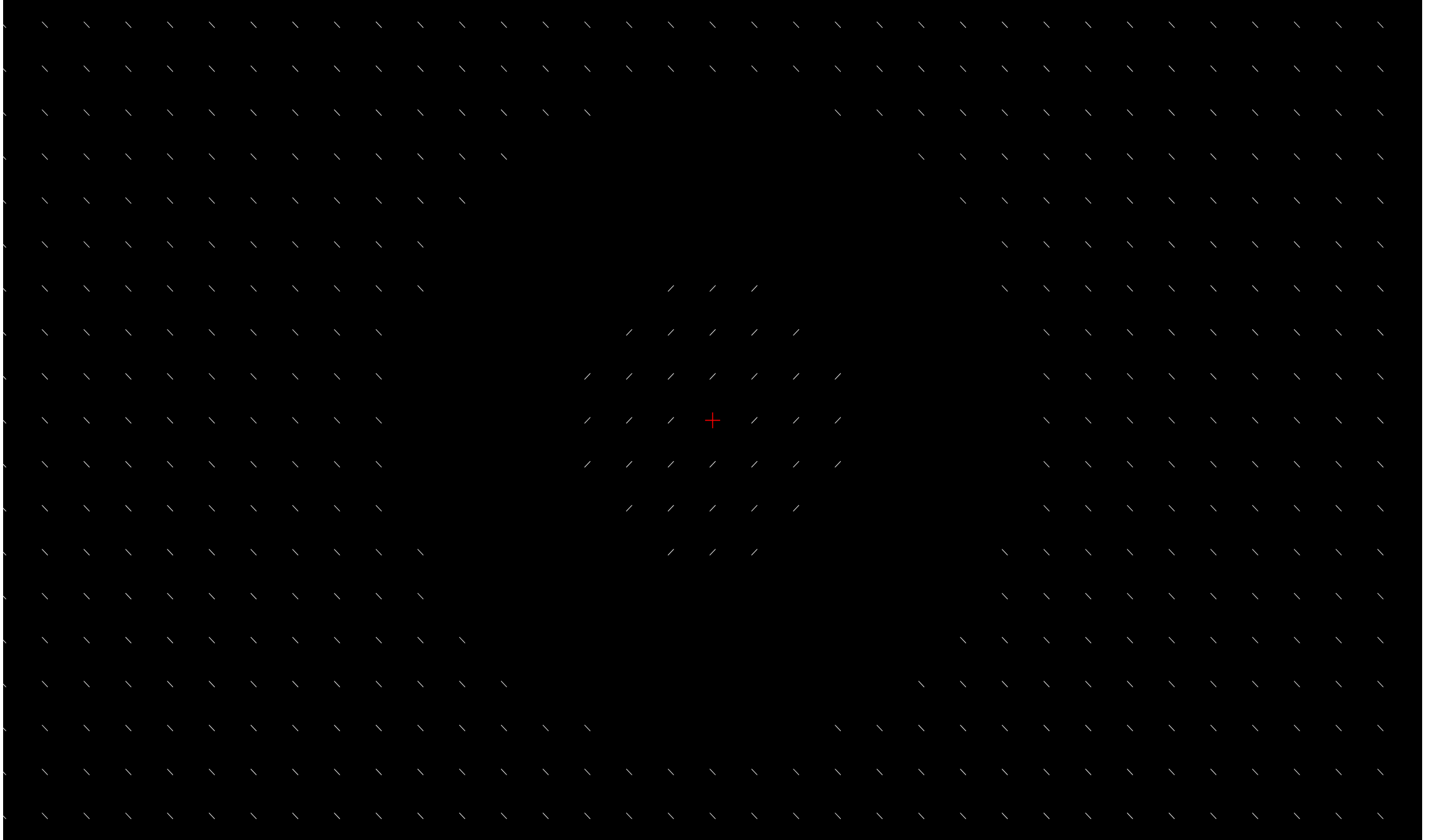


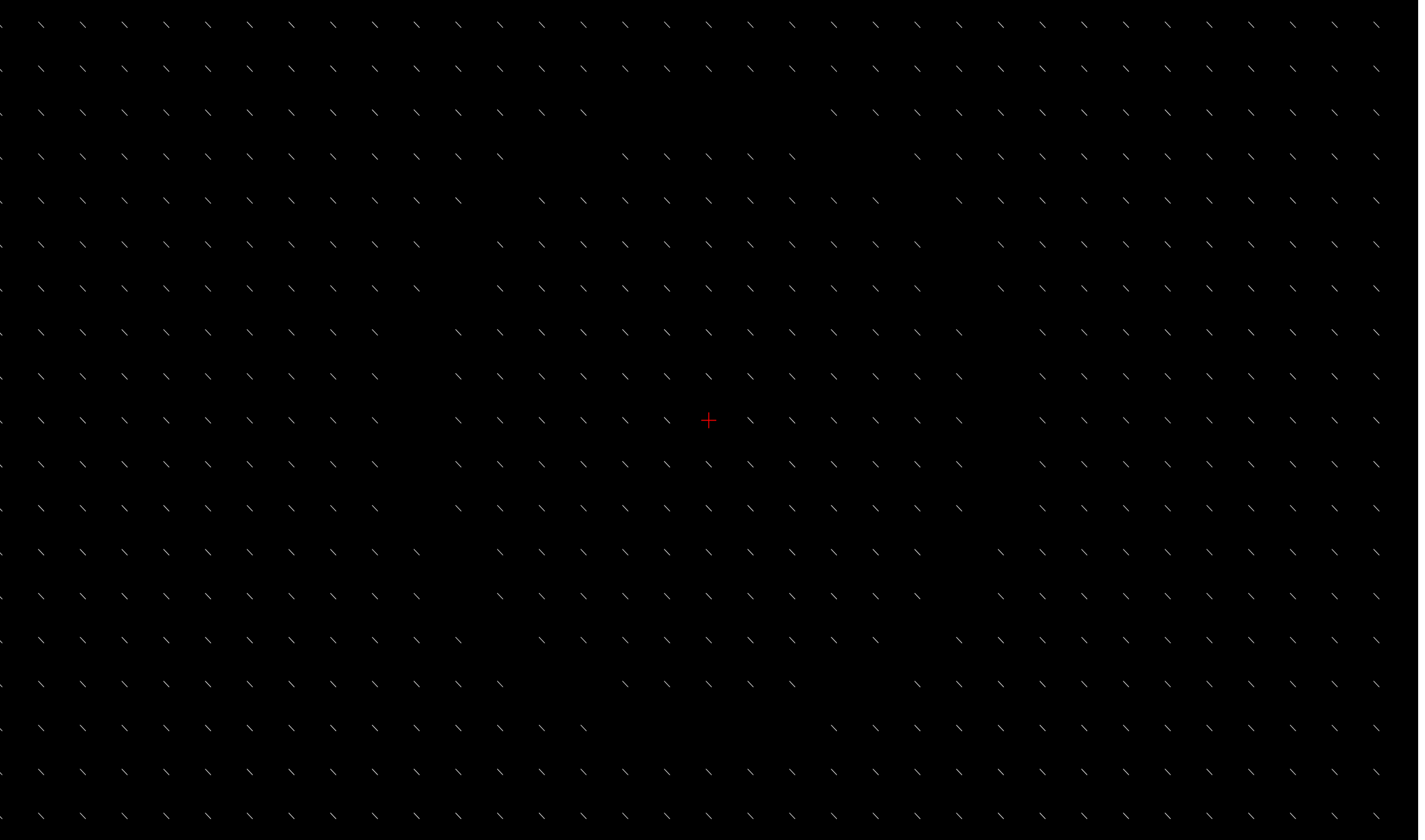
ISI (500-1000 ms)

Task: targets (peripheral) lines orientation judgement left vs right.

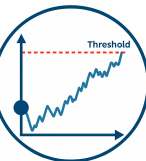


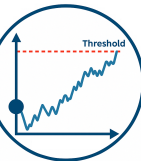
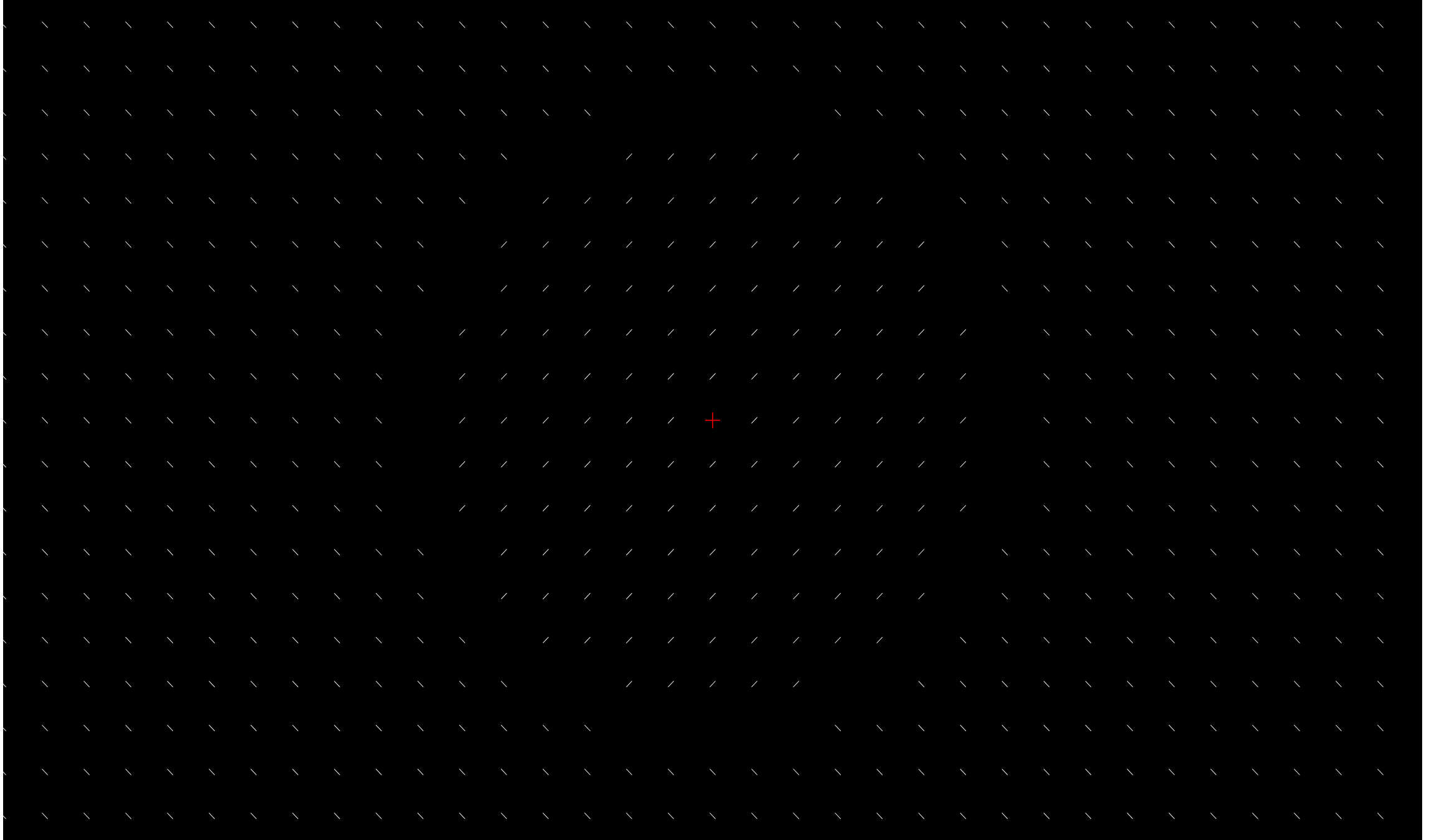






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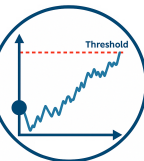




Design

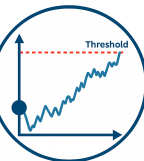
2 x 2 x 2:

- 2 **gap sizes** (small vs large)
- 2 **conditions** (congruent vs incongruent)
- 2 peripheral orientations (left vs right)



Hypotheses

1. Gap effect: **small gap** higher accuracy but slower RT (**higher threshold**)
2. Congruency effect: **incongruent conditions** have higher accuracy and faster RT (**higher drift rate**)



Let's do it!

